

National University of Singapore

School of Computing

MID-SEMESTER TEST FOR Semester 2 AY2014/2015

CS1010E — Programming Methodology

14 March 2015

Time Allowed: 60 Minutes

INSTRUCTIONS TO CANDIDATES

1. This test paper contains TWENTY (20) questions and comprises TEN (10) printed pages, including this page.
2. Every question is worth one mark.
The maximum possible mark in total is 20.
3. Answer ALL questions by shading the letter corresponding to the *most appropriate* answer on the OCR form provided.
4. This is an OPEN BOOK test.
5. Do not look at the questions until you are told to do so.
6. There is no negative marking, so please attempt every question.
7. You may keep the question paper after the test is over.

For all the questions, assume that the relevant `#include` pre-processor statements have already been included in the program where necessary. Choose the most appropriate answer for each question.

1. Which of the following, if any, is *not* a valid identifier?

- A. `Abc012`
- B. `Abc_012`
- C. `_abc012`
- D. `abc-012`
- E. All of the above are valid identifiers.

2. What is printed by the following C program fragment?

```
int a, c = 4;
double b = 8.5, d;
a = b / c;
d = 3 / a;
printf("%f\n", d);
```

- A. 0.000000
- B. 1.000000
- C. 1.500000
- D. 2.000000
- E. None of the above

3. What is printed by the following C program fragment?

```
int a, c = 4;
double b = 8.5, d;
a = b / c;
d = 3 / (double) a;
printf("%f\n", d);
```

- A. 0.000000
- B. 1.000000
- C. 1.500000
- D. 2.000000
- E. None of the above

4. Assume the `int` variable `num` is assigned a number between 5 and 50, inclusive. Which of the following expressions evaluates to *true* (ie. a nonzero value) when `num` is a positive *odd* number?

A. `num % 2 ? 0 : 1`
B. `(num - 1) % 2`
C. `(num + 1) % 2`
D. `(num + 3) % (num + 1) % 2`
E. None of the above

5. What is printed by the following C program fragment?

```
int c = 0, a = 1, b = 2;
printf("%d, ++c = a + b++);
```

A. 3
B. 4
C. 5
D. A syntax/compilation error occurs
E. A runtime error occurs

6. What is printed by the following C program fragment?

```
int x = 0;
printf("%d ", ++x);
printf("%d ", x++);
printf("%d\n", ++x);
```

A. 0 0 0
B. 0 1 2
C. 1 2 3
D. 1 1 3
E. None of the above

7. What is the value of y after the following C program fragment is executed?

```
int x = 5, y = 2;
if (x <= 4) ;
    y++;
y++;
```

- A. 2
- B. 3
- C. 4
- D. A syntax/compilation error occurs
- E. None of the above

8. What is the value of y after the following C program fragment is executed?

```
int x = 7, y = 2;
if (x < 3)
if (y > 4)
if (x < 5)
y++;
else
y--;
```

- A. 1
- B. 2
- C. 3
- D. A syntax/compilation error occurs
- E. None of the above

9. What is printed by the following C program fragment?

```
int false = -1, true = !false;
if (!true) printf("%d\n", false = -2);
else printf("%d\n", false == -1);
```

- A. -1
- B. 0
- C. 1
- D. -2
- E. None of the above

10. What is the value of c after the following C program fragment is executed?

```
int a = 1, b, c;  
c = (a > 0 ? b = ++a + (a > 2 ? 88 : 99) : 77);
```

- A. 79
- B. 90
- C. 101
- D. A syntax/compilation error occurs
- E. None of the above

11. What is the value of j after the following C program fragment is executed?

```
int i = 1, j = 0;  
switch (++i) {  
    case 1: j += i;  
    case 2: j += i; break;  
    case 3: j += i;  
    default: j += i; }
```

- A. 0
- B. 1
- C. 2
- D. 3
- E. None of the above

12. What is the value of b after the following C program fragment is executed?

```
int a = 0, b = 0;  
while (b - a < 1000) {  
    a++; b += 2;  
}
```

- A. 1998
- B. 2000
- C. 2002
- D. A syntax/compilation error occurs
- E. None of the above

13. What is the value of a after the following C program fragment is executed?

```
int a = 0, b = 999;
while (a < b || a / b > 1) {
    if (a % 2) {
        a++; b--;
    } else continue;
}
```

- A. 999
- B. 1000
- C. 1001
- D. A runtime error (divide by zero) occurs
- E. An infinite loop occurs

14. What is the value of j after the following C program fragment is executed?

```
int i = 8, j = 0;
do {
    --i;
    if (i % 2) continue;
    j++;
} while (i);
```

- A. 8
- B. 6
- C. 4
- D. 2
- E. None of the above

15. What is the value of k after the following C program fragment is executed?

```
int i, j, k = 0;
for (i = 1; i <= 5; i += 2)
    for (j = i; j >= 1; j--) k++;
```

- A. 6
- B. 7
- C. 8
- D. 9
- E. None of the above

16. What is printed by the following C program fragment?

```
int f(int, int);

main() {
    int i = 1, j = 2, k;
    k = f(i, j);
    printf("%d\n", i + j + k);
}

int f(int i, int j) {
    i++; j++;
    return i+j;
}
```

- A. 6
- B. 7
- C. 8
- D. 9
- E. None of the above

17. What is printed by the following C program fragment?

```
int f(int), g(int);

main() {
    printf("%d\n", f(2) + g(3));
}

int f(int x) { return g(5 - x); }
int g(int x) { return 3 * x; }
```

- A. 3
- B. 5
- C. 9
- D. 18
- E. None of the above

18. What is printed by the following C program fragment?

```
int myfunc();
int a = 1, b = 2;

main() {
    printf("%d\n", myfunc() + myfunc());
}

int myfunc() {
    int a = 4;
    static int b = 3;
    return (++a) + (++b);
}
```

- A. 16
- B. 17
- C. 18
- D. 19
- E. None of the above

19. What is printed by the following C program fragment?

```
int myfunc(int);

main() {
    printf("%d ", myfunc(4));
}

int myfunc(int x) {
    if (x <= 0) return 0;
    if (x % 2) {
        printf("%d ", x);
        myfunc(x - 1);
    } else {
        myfunc(x - 1);
        printf("%d ", x);
    }
}
```

- A.** 4 3 2 1 0
- B.** 0 3 4 1 2
- C.** 0 4 2 3 1
- D.** 3 1 2 4 0
- E.** None of the above

20. What is printed by the following C program fragment?

```
int f(int);
int g(int);

main() {
    printf("%d\n", g(f(3) - 2));
}

int f(int x) {
    if (x <= 0) return 0;
    return x * g(x - 1);
}

int g(int x) {
    if (x <= 0) return 0;
    return x + f(x - 1);
}
```

- A. 8
- B. 9
- C. 10
- D. 11
- E. None of the above

END of PAPER